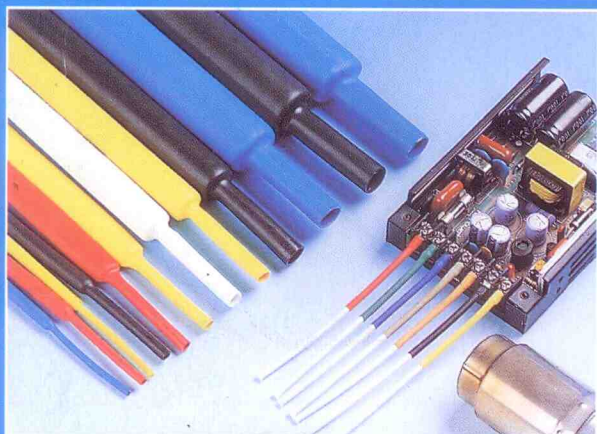


JAIN POLYMER CO.



**MARKED ADVANTAGE
MATCHLESS RANGE**

PTFE
SLEEVES & WIRES

JAIN POLYMER CO.

PTFE SLEEVES & WIRES



A few minutes delay in attacking a fire in a building can make the difference between a minor incident and a major catastrophe. Majority of fires in buildings are caused by short circuit in faulty wiring due to overload.

Conventional PVC insulated wiring succumbs to overload conditions, catches fire and emits large amount of toxic, acidic and corrosive fumes which inflict significant damage upon the surroundings. These fumes in most cases are fatal.

The Permanent Solution

Teflon insulated wires and cables primarily used in Military and aerospace applications are the answer to your wiring problems. Jain Polymers TEFLON insulated wires and cables check the occurrence of fire and resist damage like no other wires and cables available which are insulated with PVC. Teflon insulated high temperature Annealed Bare Copper wires are rated at 300°C which is 285% higher than the best FRLS (Flame Retardant Low Smoke) PVC compound insulated wires which are rated at 105°C. Conventional PVC wiring is rated at 70°C.



TEFLON WIRES AND CABLES FROM JAIN POLYMERS

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Advantages of PTFE/TEFLON Insulated Heat Proof Cables

PTFE (Polytetra Fluoro Ethylene) is a polymer material, well known these days.

It has outstanding electrical, thermal and chemical properties.

The operating temperature of PTFE is from - 65°C to +200°C for Silver plated and - 65°C to +260°C for Nickel plated wires. Type ET (250 V AC,RMS), Type E (600 V AC,RMS) and Type EE (1000 V AC,RMS) are manufactured strictly as per the specifications of JSS 51004.

Jain Polymers is a privately held company manufacturing P.T.F.E. (TEFLON) insulated Hook-Up wires, Multicore Cables, Co-axial Cables and Sleeves etc. according to the following specifications or customer specifications.

	USMIL	JSS
Equipment Wires	MIL-W-16878E	51034
Co-Axial Cables	MIL-C-17	51100
Sleeves	MIL-I-22129	54802

The advantages to TEFLON over other conventional materials

1. Excellent Thermal Stability : Most importantly TEFLON wires and cables have a wide range of service temperature ranging from - 65°C to +260°C. They withstand overheating due to transient current overload. This greatly reduces the hazards of flashover in tightly bundled wiring.
2. Fire Proof :Teflon does not propagate flame or serve as fuel for fire. It is self extinguishing and non toxic.
3. Wide Frequency : Suitable for very wide frequency range (D.C. to above 10,000 MHz) over wide temperature range.
4. Good mechanical strength and flexibility.
5. Low dielectric constant and dissipation factor not altered by heat aging.
6. Due to Teflon's high insulating properties, Teflon wires and cables are compact compared to conventional materials therefore more circuits per conduit are possible.
7. Teflon insulated wires and cables are unaffected by long term exposure to virtually all chemical and Corrosive environments including acids, bases, paints, lacquers and fuels etc.

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8. Teflon is resistant to U.V. radiation and stress cracking. It resists damage from abrasion during conduit pulling or rough handling. Teflon has exhibited no significant changes in its electrical, mechanical, chemical or thermal properties after nearly 30 years of outdoor exposure at DuPont's test areas at Florida and Michigan.
9. Immune to fungus and water absorption.
10. Soldering iron resistant and suitable for high density wiring.

For wire and cable construction, conductors with 1, 7, 19, 37 and 133 strands are available as standard sizes. Cables are available as single core, multicore and multipair twisted. They can be further shielded, Double shielded or flat woven. The choice of conductors being silver plated Copper, Silver plated Copper clad steel and Nickel-plated Copper. A choice of 12 colours namely, Black, Brown, Red, Orange, Yellow, Green, Blue, Violet, Grey, White, Pink, Natural and Bi-Colours are available for wires and cables.



MIL-W-16878	VOLTAGE RATING AC RMS	TESTING	
		DIELECTRIC	SPARK
ET	250V	1500V	2500V
E	600V	2000V	3400V
EE	1000V	3000V	5000V

What determines the ampacity of wire ?

The ampacity of any given size of wire varies with the kind of insulation on the wire and the method of installation. Copper is not harmed by heat; insulation is harmed by heat, If insulation is overheated, it is harmed in various ways depending upon the degree of overheating and the kind of insulation. Some kinds melt, some harden, some burn. In any event insulation loses its usefulness if overheated, leading to breakdowns and fires.

TEFLON WIRES AND CABLES FROM JAIN POLYMERS

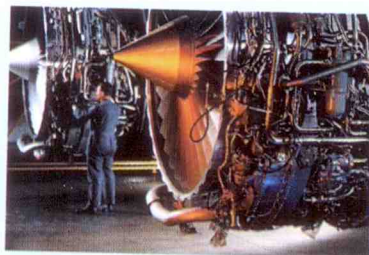
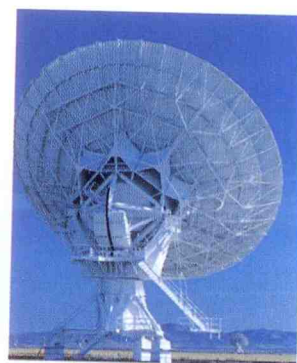
JAIN POLYMER CO.

PTFE SLEEVES & WIRES

APPLICATIONS

Our Stringent quality control procedures have enabled us to develop zero-defect products used in a variety of electronic industries across the country. Some of the application areas are :

- * Computers.
- * Refrigeration equipments.
- * Electronic Test Equipments.
- * Communication Equipments (mobile and wireless sets).
- * Electronic Equipment for Aircraft, Radar, Navigation, Electra-Mechanical application.
- * Flight Simulators.
- * Electronic Control Equipments for Atomic Energy, Satellite launching, Ground Control, Reactor and Process Control, Telephone Exchange, Lifts.
- * High performance motors, transformers and rectifiers.
- * High temperature control valves, Furnance and oven wiring.
- * Load Cells and pressure transducers.
- * Infra Red sensing equipments.
- * Air field lighting equipments.
- * Thermocouple and heat sensing leads.
- * Control panel wiring.



DOMESTIC AND INDUSTRIAL PERMANENT SOLUTIONS TO YOUR WIRING PROBLEMS

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Specifications of PTFE wires

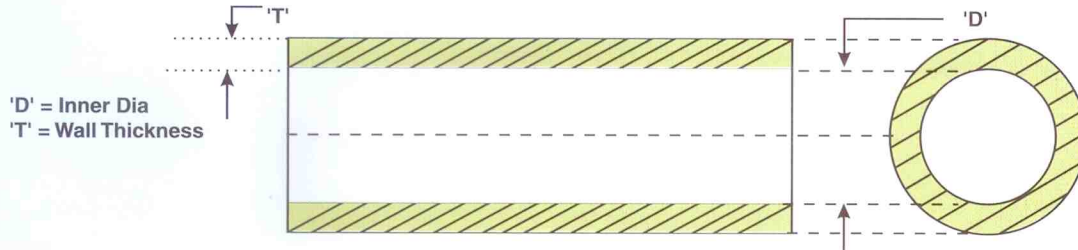
Size	Number of Strands, Dia. of each Strand		Dia. of Conductor (mm)	Cross Sectional Area (mm)	ET 250 V		E600 V		EE 1000 V		Max Resistance of Conductor OHMS/1000 mtrs. at 20°C	Current Rating (amps)
					Dia. over Insulation		Dia. over Insulation		Dia. over Insulation			
	AWG	AWG			(mm)	Min(mm)	Max(mm)	Min(mm)	Max(mm)	Min(mm)		
32	7/40	7/0.08	0.25	0.035	0.50	.060	0.66	0.86	0.91	1.11	567.57	0.7
30	1/30	1/0.25	0.25	0.049	0.50	0.60	0.66	0.86	0.91	1.11	354.32	1.0
30	7/38	7/0.10	0.31	0.055	0.55	0.66	0.71	0.91	0.96	1.16	330.37	1.0
28	1/28	1/0.32	0.32	0.080	0.58	0.68	0.73	0.93	0.99	1.19	223.09	2.1
28	7/36	7/0.13	0.38	0.093	0.63	0.73	0.78	0.99	1.04	1.24	209.31	2.1
26	1/26	1/0.40	0.40	0.125	0.66	0.76	0.81	1.01	1.06	1.27	140.09	3.0
26	7/34	7/0.16	0.48	0.140	0.73	0.83	0.88	1.09	1.14	1.34	132.87	3.0
26	19/38	19/0.10	0.51	0.149	0.73	0.83	0.88	1.09	1.14	1.34	125.98	3.0
24	1/24	1/0.51	0.51	0.204	0.76	0.86	0.91	1.11	1.16	1.37	87.92	4.0
24	7/32	7/0.20	0.61	0.219	0.86	0.96	1.01	1.21	1.27	1.47	82.67	4.0
24	19/36	19/0.13	0.64	0.252	0.86	0.96	1.01	1.21	1.27	1.47	79.72	4.0
22	1/22	1/0.64	0.64	0.321	0.88	1.01	1.04	1.27	1.29	1.52	55.77	7.3
22	7/30	7/0.25	0.76	0.343	1.01	1.11	1.16	1.37	1.42	1.62	52.16	7.3
22	19/34	19/0.16	0.81	0.382	1.01	1.11	1.16	1.37	1.42	1.52	49.543	7.3
20	1/20	1/0.81	0.81	0.515	1.06	1.16	1.21	1.42	1.47	1.67	34.44	11.0
20	7/28	7/0.32	0.97	0.562	1.21	1.32	1.37	1.57	1.62	1.82	32.80	11.0
20	19/32	19/0.20	1.02	0.596	1.21	1.32	1.37	1.57	1.62	1.82	30.15	11.0
18	7/26	7/0.40	1.22	0.879			1.63	1.87	1.87	2.13	20.60	16.0
18	19/30	19/0.25	1.27	0.932			1.63	1.87	1.87	2.13	18.99	16.0
16	19/29	19/0.29	1.45	1.254			1.85	2.20	2.10	2.41	14.82	22.0
15	19/28	19/0.32	1.60	1.528			1.98	2.28	2.26	2.54	12.09	26.0
14	19/27	19/0.36	1.83	1.933			2.23	2.59	2.48	2.89	9.44	32.0
13	19/26	19/0.40	2.00	2.387			2.43	2.74	2.66	3.02	7.59	35.0
12	19/25	19/0.45	2.31	3.021			2.71	3.07	2.97	3.37	5.93	41.0
12	37/28	37/0.32	2.26	2.975			2.66	3.02	2.92	3.32	6.23	41.0
11	19/24	19/0.50	2.50	3.730			2.99	3.30	3.20	3.58	4.72	45.0
10	19/22	19/0.65	3.25	6.304			3.73	4.03	3.96	4.34	2.99	55.0
10	37/26	37/0.4	2.82	4.649			3.22	3.58	3.47	3.88	3.51	50.0
8	133/29	133/0.29	4.29	8.784					5.05	5.56	2.15	75.0
6	133/27	133/0.36	5.41	13.537					6.42	6.93	1.37	100.0

TEFLON WIRES AND CABLES FROM JAIN POLYMERS

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PTFE SLEEVES & WIRES

P.T.F.E. Sleeves



Intended Use - The sleeving covered by this specification is intended for applications involving high temperature, high frequencies or both. Sleeving covered by this specification possesses excellent electrical properties which remain practically constant over a wide range of frequencies, temperatures, and humidities and has a very low loss factor.

Specifications of PTFE Sleeves

S. No.	BORE		WALL THICKNESS		BREAKDOWN VOLTAGE
	Nominal (mm)'D'	Tolerance (mm)	Nominal (mm)'T'	Tolerance (mm)	BDV, AC RMS K.Volts
1.	0.50	±0.10	0.25	±0.05	11.5
2.	0.75	±0.10	0.25	±0.05	11.5
3.	1.00	±0.10	0.30	±0.08	14.6
4.	1.50	±0.15	0.40	±0.08	16.3
5.	2.00	±0.15	0.40	±0.08	16.3
6.	2.50	±0.15	0.40	±0.08	16.3
7.	3.00	±0.15	0.40	±0.08	16.3
8.	3.50	±0.20	0.5	±0.1	17.0
9.	4.00	±0.20	0.5	±0.1	17.0
10.	5.00	±0.20	0.5	±0.1	17.0
11.	6.00	±0.30	0.5	±0.1	17.0
12.	7.00	±0.30	0.5	±0.1	17.0
13.	8.00	±0.30	0.5	±0.1	17.0
14.	9.00	±0.30	0.5	±0.1	17.0

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Choice of Sleeving

Although many wire-ended capacitors and resistors are used on tag boards without sleeving, this is not so when point-to-point wiring is necessary, as in, for example, a miniature I.F. strip.

In this, and like instances, the use of sleeving on bare wire is generally a quick and effective wiring method. It should be borne in mind that there will be a space between the sleeving and conductor which is not present with covered wire.

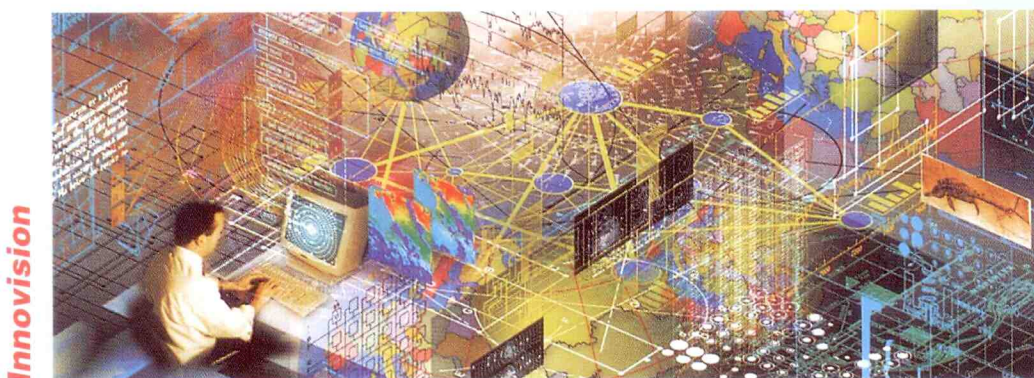
Moisture may enter into this space and become trapped, causing subsequent corrosion of the conductor and a lowering of insulation resistance when the sleeved wires touch. In a similar manner large diameter sleeving, when used to hold together a number of covered wires, may form a dangerous moisture trap.

In all cases sleeveings should be chosen to be as tight a fit as possible on the conductor and, to facilitate insertion, the use of a suitable lubricant may be advantageous.

* TEFLON is a registered trademark of DUPONT

ORDERING INSTRUCTIONS

1. Please specify wall thickness and colour. All standards colours available, others on request.
2. Minimum order 100m of one size and colour.
3. Standard packing is of 50m coils (1 to 5m Nominal piece length) and 50m bundles for sizes above 3mm bore (1mtr. nominal length)



" Innovation with vision for future, is the prime focus of our organisation."



JAIN POLYMER CO.

Manufacturers of :
PTFE/TEFLON WIRE, CABLE, SLEEVE & TAPE

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